StepAhead

BUILD PLAN + LEARNING PLAN

Youth Investing Simulator

A paper trading app on real stock and forex data that teaches young people compound interest, risk and diversification. Measured by one thing: how many people learn to invest responsibly with it.

STEPAHEAD · START HERE

Start here, Arjun

You only do this setup once. After that, Claude Code remembers the plan, remembers where you stopped, and coaches you through every step. You never have to work out what's next.

1 **Open Terminal**

Mac: press Cmd + Space, type Terminal, press Enter. Windows: open PowerShell.

2 **Make the project folder and start Claude Code**

Paste this, press Enter:

```
mkdir -p ~/youth-investing-simulator && cd ~/youth-investing-simulator && claude
```

If it says `claude: command not found`, run `npm install -g @anthropic-ai/claude-code` and try again.

3 **Give it this document**

Drag this PDF into the Claude Code window, type `start`, press Enter. That's it. Claude takes it from here.

Every time after that

↻ **Two things, any day, any time**

```
cd ~/youth-investing-simulator && claude
```

Then type `continue`. Claude reads where you stopped and picks up from there, even if it's been a while.

What a session feels like

- **30 to 45 minutes.** One small piece of a milestone. Short sessions often beat long ones once in a while.
- **Claude writes the code, you run the important commands** yourself, so you learn what they do.
- **Every new idea is explained in one line** and saved to your own notes file, `LEARNED.md`. By the end it's your personal guide to building and investing.
- **You make one real choice every milestone,** like a colour, a chart or a lesson topic. It's your app.
- **Errors are normal.** Every developer sees them daily. Claude shows you how to read them.
- **You always end on a win:** a two line recap of what you can now do, and what next time starts with.

Watch it grow

Your site shows a progress strip from M1 to M16, your impact level, and a Build log of everything you've added. Open your link anytime and you'll see how far you've come.

STEPAHEAD · BUILD PLAN

Youth Investing Simulator

PREPARED BY	Sahil
BUILDER	Arjun
GOAL	Build a paper trading app on real market data that teaches compound interest, risk and diversification, ship it on one link, and get it used by real young learners, with the learning measured.
DATA	Frankfurter (forex, free, no key) and Twelve Data (stocks, free key, no card)
STACK	GitHub, GitHub Pages, GitHub Actions, Next.js on Vercel, Supabase, Google sign in. All free plans.

How the milestones work

One link, always

The link from M1 is the only link Arjun shares. It never changes; it just gets better.

Update after every milestone

Same link, plus what was added and what was learned. Sent to Sahil on WhatsApp.

Judged by what works

A milestone is done when Sahil can check the "Done when" line on the live link. Not by hours spent.

The community impact scale

The app is the means. The outcome is learners. A learner counts once they have finished at least one lesson and one simulation.

LEVEL	NAME	WHAT IT MEANS	REACHED AT
Level 0	**Built**	It works for Arjun	M1 to M10
Level 1	**Friends**	10 learners finish a lesson and a simulation	M13
Level 2	**Classroom**	30 learners, quiz improvement measured	M14
Level 3	**School**	100+ learners, a teacher uses it on their own	M16
Level 4	**Community**	500+ learners across several schools or groups	Next phase

Milestone update

```
Milestone N done
Link: (the one link)
New: (2 or 3 things anyone can try)
Learned: (1 or 2 things)
Learners so far: (from the impact page, from M12)
Next: Milestone N+1
```

Nothing shipped this week? Say so in one line, and say why. That's fine; silence isn't.

STEPAHEAD · FOR CLAUDE CODE

For Claude Code

Arjun, you can skip this page. It tells Claude Code how to run your plan.

Claude Code: this document is the whole plan. When Arjun types `start`, do the first session setup below, once. In every session after, follow the `CLAUDE.md` you created.

First session, once

1. Ask, then wait: Mac or Windows? First name? GitHub username? (If none, walk him through making one.)
2. Check git, node and gh. Install anything missing, one at a time, checking each works.
3. Save this PDF to `docs/plan.pdf`. Write every milestone (goal, deliverables, Done when, Learn) into `PLAN.md` word for word, plus the impact scale and the rules on the overview page.
4. Create `CLAUDE.md` with exactly the text below (put in his username), `PROGRESS.md` set to *M1, part 1*, and an empty `LEARNED.md`.
5. Commit, explain in one line what these files are for, then begin M1.

```
# Youth Investing Simulator: how we work

Builder: Arjun, a beginner. You are his coach and pair programmer.
The plan is PLAN.md (docs/plan.pdf is the original). Progress is PROGRESS.md. His notes are LEARNED.md.

## Every session
1. Read PROGRESS.md and `git log -5`. Greet Arjun in one line with where we are and what today's piece is.
2. If the last commit is more than 7 days old, welcome him back warmly, no guilt, and start with a 15 minute win.
3. Work only on the current milestone. Split it into parts of 30 to 45 minutes. One part is a good session.
4. Start each milestone with one line on why it matters in the real world.

## How to coach
- Short replies: one step, one or two lines. Wait for him before the next step.
- You write the code. Arjun runs the important commands himself (git, gh, npm, vercel, supabase)
  by typing ! before them. Give him the exact command and say in one line what it does.
- Explain every new idea in one plain sentence, then add one line to LEARNED.md in simple words.
- Let him make one real choice per milestone (a colour, a name, a chart, a lesson topic).
- Errors are normal. Show him how to read one, then fix it together.
- Never ask him to edit files by hand. Use git commit -m so no editor opens.
- If he seems stuck or tired, make the step smaller. Celebrate every part finished.

## Rules
- Free plans only. Nothing that needs a credit card. No paid AI APIs (no Anthropic or OpenAI keys).
- Keys only in .env.local, GitHub Secrets or Vercel settings. Run git status before every commit; never push a key.
- One link forever: https://USERNAME.github.io/youth-investing-simulator. From M7 it forwards to the Vercel app.
- Pretend money only. "Educational, pretend money, not financial advice" on every page.
- The site has a progress strip (M1 to M16 plus current impact level) and a Build log. Keep both current.
- Follow PLAN.md. Don't skip milestones or add big features that aren't in it.

## End of every session
- Commit and push. Update PROGRESS.md: what's done, the current part, the exact next step.
- Give Arjun a 2 line recap: what he can now do himself, and what next session starts with.

## End of every milestone
1. Check the "Done when" line on the live link. If it fails, fix it first.
2. Add a Build log entry and move the progress strip.
3. Ask him one quick question from LEARNED.md to check it stuck.
4. Write his update in a box to copy, and tell him to send it to Sahil on WhatsApp:
   Milestone N done / Link / New / Learned / Learners so far / Next
```

STEPAHEAD · BUILD PLAN

STAGE 1 Foundations

A live link, the maths of money, and real market data flowing in by itself.

M1 Repo and live link

Get the project on the internet on day one, with one link that never changes.

- GitHub repo `youth-investing-simulator` with a README that says who it is for and why
- A landing page on GitHub Pages: name, one line mission, and a Build log section
- Works on a phone

Done when: Sahil opens the link on his phone. This is the link Arjun shares at every milestone from now on.

Learn: terminal, git, GitHub, how a repo becomes a website

M2 Compound interest calculator

The first thing every young investor should feel: time matters more than timing.

- Inputs: starting amount, monthly top up, yearly return, years
- A growth chart, and a side by side line showing what starting 5 years later costs
- The maths lives in its own file, separate from the page, with tests

Done when: ₹1,000 at 8% a year for 10 years shows ₹2,158.92. Three test cases pass from the terminal.

Learn: functions, the compound interest formula, separating logic from UI, Chart.js

M3 Real market data, refreshed by itself

Real prices, free, updated every night without Arjun touching anything.

- Forex from Frankfurter (free, no key): USD, EUR, GBP against INR
- Daily prices for a starter list of about 20 well known stocks and index funds from Twelve Data's free plan (free key, no card)
- A GitHub Action that runs nightly and saves the prices as JSON; every page shows *Data as of <date>*
- The API key lives in GitHub Secrets, never in the code

Done when: The Action runs on its own for 3 nights in a row and the date on the site moves forward.

Learn: APIs, JSON, scheduled jobs, keeping a key secret

STAGE 2 The simulator

Paper trading that teaches, not paper trading that gambles.

M4 Paper trading core

Every learner starts with ₹1,00,000 of pretend money.

- Buy and sell at the latest closing price; holdings, cash and total value always visible
- Trade history with date, price and quantity
- Saved in the browser so it survives a refresh
- Simple rules on purpose: no borrowing, no short selling, no leverage

Done when: Sahil makes 20 trades with no glitches, and cash plus holdings always equals total value.

Learn: state, edge cases (selling more than you own, spending more than you have)

M5 Risk and diversification

Show learners what their choices actually mean.

- Portfolio split by asset, sector and currency as a chart
- Risk shown as how much the portfolio moves day to day (volatility), in plain words
- A diversification score from 0 to 100, with a gentle nudge like *70% of your money is in one stock*

Done when: Three test portfolios (one stock, five stocks, a mix with an index fund) get scores that make sense and the same score every time.

Learn: averages, standard deviation, explaining a number to a non expert

M6 Learn mode

Five short lessons, each under five minutes, each ending in a quiz.

- Compound interest · Risk · Diversification · How forex works · Investing vs trading
- A short starter quiz before the first lesson and the same quiz after the last one
- Lessons link straight into the simulator: *now try it*

Done when: A friend with no finance background finishes lesson 1 in under 5 minutes and gets the quiz right.

Learn: writing for beginners, measuring learning with a before and after

STEPAHEAD · BUILD PLAN

STAGE 3 Real app

Accounts, a database and challenges, on free plans. The link from M1 keeps working.

M7 Move to a real app

Same product, now on the stack real startups use.

- Rebuilt in Next.js and hosted on Vercel (free Hobby plan)
- The GitHub Pages link forwards to the new app
- Every push to GitHub creates a new version automatically

Done when: The old link opens the new app, and every feature from M1 to M6 still works.

Learn: frameworks, deployments, previews

M8 Accounts with Google sign in

Learners keep their portfolio and progress on any device.

- Supabase (free plan) for the database and sign in
- Sign in with Google through Supabase Auth
- Portfolios, trades and quiz results saved per learner, with row level security so nobody sees anyone else's
- Keys in `.env.local` and Vercel settings only

Done when: A friend signs in, makes trades, and cannot see Arjun's portfolio. No key appears anywhere on GitHub.

Learn: databases, SQL basics, OAuth, data privacy

M9 Time machine and weekly challenge

Learn from history without waiting years.

- Time machine: *invest ₹10,000 in 2016, see where it is today*, with and without diversification
- Weekly challenge: the same starting scenario for everyone, resets every Monday

Done when: Sahil runs the time machine for 3 portfolios and the results match the real price history.

Learn: historical data, seeded scenarios

M10 A leaderboard that rewards being responsible

Rank learners on smart investing, not lucky bets.

- Score = return adjusted for risk, plus diversification, plus lessons completed
- Display names only, no real names or photos
- A short *why this ranking* note so learners see what is being rewarded

Done when: A portfolio that put everything in one lucky stock does not top the board.

Learn: designing incentives, ethics of ranking

STAGE 4 Community impact

The outcome is not the app. It is how many young people learn to invest responsibly with it.

M11 Safe for young learners

Earn the trust of parents and teachers before reaching them.

- A clear line on every page: *educational, pretend money, not financial advice*
- A privacy page in plain words: what is stored, what is never stored, how to delete an account
- Only the data the app needs; no phone numbers, no real money, no bank links

Done when: A parent reads the privacy page and understands it in two minutes.

Learn: data minimisation, writing policy in plain language

M12 Public impact page

Show the outcome, not just the features.

- A public /impact page with anonymous totals: learners, lessons completed, simulations run
- Average quiz improvement, before vs after (from M6)
- Updates by itself from the database

Done when: The numbers on /impact match the database, and no individual can be identified.

Learn: aggregate queries, honest metrics

M13 Pilot with 10 friends

Real users before more features.

- 10 friends sign up and complete at least one lesson and one simulation
- A free Google Form for feedback
- Fix the top 3 problems they hit

Done when: Impact Level 1 reached, and Arjun can name the top 3 fixes and why he chose them.

Learn: user testing, prioritising

STAGE 4 Community impact (continued)

From one class to many schools.

M14 Classroom pack

Make it easy for a teacher to use with a whole class.

- Class codes: a teacher creates a class, learners join with a code
- A teacher view with class progress and quiz improvement
- A one page teacher guide

Done when: One class or club of about 30 uses it in a session. Impact Level 2.

Learn: roles and permissions, building for a second kind of user

M15 Release v1.0 safely

Ship updates without fear.

- Version tags and short release notes for every release
- A before and after release checklist
- A known good version to roll back to on Vercel

Done when: Sahil can see what changed in each version, and one rollback has been tested.

Learn: versioning, release discipline

M16 Grow the reach

Take it beyond one school.

- Share with at least 3 schools, clubs or youth groups
- A one page impact report: learners reached, quiz improvement, what learners said
- A plan for the next 500 learners

Done when: Impact Level 3: 100+ learners, with the improvement measured.

Learn: presenting impact, partnerships

What you'll learn while building

- **Money:** compound interest, risk, diversification, how currencies move, investing vs trading
- **Terminal and git:** working from the terminal, committing every step, releasing with tags
- **Data:** calling real APIs, scheduled pipelines, cleaning and storing data
- **Product:** a web app with accounts, a database, privacy and permissions
- **Analytics:** returns, volatility, scores, before and after measurement
- **Impact:** testing with real users, working with teachers, reporting outcomes honestly

Scope notes

- Pretend money only. No real trading, no bank or broker links, ever.
- Educational, not financial advice. The app says so on every page.
- Free plans only. Nothing that needs a credit card, and no paid AI APIs.
- Daily closing prices are enough. Live tick by tick data is out of scope.
- Built for young learners, so privacy and safety come before growth.

Guiding principle

Works for me → works for a friend → works for a class → then grow. Don't grow something that doesn't work yet, and don't keep polishing something nobody is using.

When unsure what to do next

1. Does it run on the live link?
2. Can a friend use it without being told how?
3. Does it teach something they didn't know?
4. Will one small change help more people learn?

If the answer to 1 or 2 is no, fix that first.